

Week 1 Objective: Identify literature and develop Outline/Structure

Activity	Status	Literature	Who?
Review literature on datasets and algorithms used in fraud detection studies	Machine Learning Algorithms Used in Fraud Detection • Logistic Regression • Decision Trees • Random Forest • Support Vector Machine (SVM) • K-Nearest Neighbours (KNN) • Naïve Bayes • Artificial Neural Networks (ANN) • Deep Neural Networks (DNN) • Autoencoders • XGBoost • LightGBM • CatBoost • Gradient Boosting Machines (GBM) • Isolation Forest • Ensemble Learning Models • Hybrid Machine Learning Models Algorithm Evaluation Criteria • Accuracy • Precision • Recall • F1-score • ROC-AUC • False Positive Rate • Detection latency • Computational efficiency • Scalability • Explainability and interpretability	AbouGrad and Sankuru (2025); Almazroi and Ayub (2023); Alrasheedi (2025); Chen (2025); Dichev, Zarkova and Angelov (2025); Du et al. (2024); Eswar Prasad et al. (2023); Hashemi, Mirtaheri and Greco (2022); Islam et al. (2024); Khalid et al. (2024); Li and Chen (2025); Qi (2025); Zhang et al. (2023); Zhang et al. (2022)	Ifeluni

	Datasets and Datasets Characteristics - Dataset size and transaction volume - Fraud-to-legitimate transaction ratio - Feature types (transactional, behavioural, demographic) - Data quality and completeness - Availability of labelled fraud cases - Suitability for supervised and unsupervised learning - Advantages and limitations of each dataset		
Identify and evaluate publicly available fraud detection datasets	- IEEE CIS Fraud Dataset (https://www.kaggle.com/competitions/ieee-fraud-detection) (Almazroi and Ayub, 2023) - Credit Card Fraud dataset (AbouGrad and Sankuru, 2025): https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud - Bank Account Fraud Dataset Suite (NeurIPS 2022): https://www.kaggle.com/datasets/sgpjesus/bank-account-fraud-dataset-neurips-2022 - Synthetic data from a financial payment system: https://www.kaggle.com/datasets/ealaxi/banksim1/	AbouGrad and Sankuru (2025); Almazroi and Ayub (2023); Hashemi, Mirtaheri and Greco (2022); Islam et al. (2024); Zhang et al. (2023)	Ifelunni
Select the most appropriate dataset for the study	Selected Datasets - Bank Account Fraud Dataset Suite (NeurIPS 2022) - Credit Card Fraud dataset ➤ Using the Bank Account Fraud Dataset Suite (NeurIPS 2022) and the Credit Card Fraud Dataset to facilitate evaluation of the robustness and generalisability of machine learning algorithms across different fraud detection scenarios. ➤ Compare the performance of individual and ensemble machine learning models on both datasets using standard evaluation metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and false-positive rate.	AbouGrad and Sankuru (2025); Almazroi and Ayub (2023); Hashemi, Mirtaheri and Greco (2022); Islam et al. (2024); Zhang et al. (2023)	Ifelunni

	➤ Assess the consistency, scalability, and transferability of the selected models across datasets to identify the most effective algorithms for online banking fraud detection.		
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